A global model of where unrest leads to conflict: model equations

Notation

s, t	0.5° cell, week
$j \in \{\mathrm{C}, \mathrm{U}\}$	outcome: conflict, unrest
$c(s)$, $r(s)$	country and region of cell s (Africa, Middle East, South and Southeast Asia, Latin America)
$y(t)$	calendar year of week t
$g(s)$, $k(t)$	coarse cell (1° or 2°) and time step (Δ weeks) of the space-time fields
$\mathscr{T}_r$	weeks observed in region r (from the start of ACLED coverage to 2025)

Level 1: data

For $t \in \mathscr{T}_{r(s)}$,

$$Y_{jst} \mid \eta_{jst} \sim \text{Bernoulli}\big(\pi_{jst}\big), \qquad \operatorname{logit} \pi_{jst} = \eta_{jst},$$

independent over j, s and t given the linear predictors. $Y_{jst} = 1$ if at least one event of type j is recorded in cell s in week t.

Level 2: linear predictors

$$\begin{aligned}
\eta_{\mathrm{C}st} = {} & \underbrace{x_{st}^\top \beta_{\mathrm{C}} + \alpha_{\mathrm{C},r(s),y(t)}}_{\text{covariates, region-year}} + \underbrace{\delta_{\mathrm{C}}\, h_{\mathrm{C}st} + \delta_{\mathrm{C}}^{nb}\, \bar{h}_{\mathrm{C}st}}_{\text{own history}} + \underbrace{\kappa_{\mathrm{U}\to\mathrm{C}}(s)\, h_{\mathrm{U}st}}_{\text{escalation}} \\
& + \underbrace{u_{\mathrm{C},c(s)} + e_{\mathrm{C}s} + a_s + w_{\mathrm{C}s}}_{\text{static}} + \underbrace{v_{\mathrm{C},g(s),k(t)}}_{\text{space-time}}, \\
\eta_{\mathrm{U}st} = {} & x_{st}^\top \beta_{\mathrm{U}} + \alpha_{\mathrm{U},r(s),y(t)} + \delta_{\mathrm{U}}\, h_{\mathrm{U}st} + \delta_{\mathrm{U}}^{nb}\, \bar{h}_{\mathrm{U}st} + \kappa_{\mathrm{C}\to\mathrm{U}}(s)\, h_{\mathrm{C}st} \\
& + u_{\mathrm{U},c(s)} + e_{\mathrm{U}s} + \lambda_a\, a_s + w_{\mathrm{U}s} + \lambda(s)\, v_{\mathrm{C},g(s),k(t)} + v_{\mathrm{U},g(s),k(t)}.
\end{aligned}$$

Histories (observed, computed from the outcomes; N_{jst} is the number of recorded events):

$$h_{jst} = \log\Big(1 + \textstyle\sum_{t'=t-4}^{t-1} N_{jst'}\Big), \qquad \bar{h}_{jst} = \frac{1}{|R_1(s)|} \textstyle\sum_{s' \in R_1(s)} h_{js't},$$

with $R_1(s)$ the rook neighbours of s (up to four cells sharing an edge). h_{jst} is the history of the *same* cell over the four weeks before t.

Escalation acts only through the other outcome's history in the same cell over the previous four weeks, with a coefficient that varies in space:

$$\kappa_{\mathrm{U}\to\mathrm{C}}(s) = \kappa_{\mathrm{U}\to\mathrm{C}} + \gamma_{\mathrm{U}\to\mathrm{C}}(s), \qquad \kappa_{\mathrm{C}\to\mathrm{U}}(s) = \kappa_{\mathrm{C}\to\mathrm{U}} + \gamma_{\mathrm{C}\to\mathrm{U}}(s).$$

Terms for the neighbours' unrest, for longer lags (5–26 weeks) and for cells 50–300 km away are left out. The neighbour term is about zero in both fitted regions, and a diagnostic on the fitted latent fields shows no unexplained lagged association between unrest and conflict at 13- and 26-week lags (Supplementary Material).

Level 3: latent effects, given the hyperparameters θ

All terms below are mutually independent.

country:	$u_{j,c} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2_{u,j})$,
cell, outcome-specific:	$e_{js} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2_{e,j})$,
cell, shared (new):	$a_s \overset{iid}{\sim} \mathcal{N}(0, \sigma^2_a)$, loading 1 for conflict, λ_a for unrest,
static field:	$\boldsymbol{w}_j \sim \mathcal{N}\big(\mathbf{0}, \boldsymbol{Q}_s(\rho_{w,j}, \sigma_{w,j})^{-1}\big)$,
space-time fields:	$v_{j,g,k} = \varphi_j\, v_{j,g,k-1} + \sqrt{1-\varphi_j^2}\, \varepsilon_{j,g,k}$, $\quad \varepsilon_{j,\cdot,k} \sim \mathcal{N}\big(\mathbf{0}, \boldsymbol{Q}_s(\rho_j, \sigma_j)^{-1}\big)$,

with $v_{j,g,1} = \varepsilon_{j,g,1}$ (stationary start) and $\varepsilon_{j,\cdot,k}$ independent over k. $\boldsymbol{Q}_s(\rho, \sigma)$ is the finite-volume SPDE precision of a Matérn field ($\nu = 1$, range ρ, SD σ) on the sphere, with sea barriers.

Spatial variation, one of two versions:

	piecewise	smooth
escalation	$\gamma_{\mathrm{U}\to\mathrm{C}}(s) = \gamma_{\mathrm{U}\to\mathrm{C},r(s)}$, fixed effects	$\gamma_{\mathrm{U}\to\mathrm{C}} \sim \mathcal{N}\big(\mathbf{0}, \boldsymbol{Q}_s(\rho_\gamma, \sigma_\gamma)^{-1}\big)$
loading	$\lambda(s) = \lambda_{r(s)}$	$\lambda(s) = \sum_{m=1}^{M} \vartheta_m\, b_m(s)$, $M \approx 5$–10

Likewise for $\gamma_{\mathrm{C}\to\mathrm{U}}$. The loadings λ_a and $\lambda(s)$ enter the prior precision of the pairs $(a_s, \lambda_a a_s)$ and $(v_{\mathrm{C}}, \lambda v_{\mathrm{C}})$ (copy construction), whose log-determinant does not depend on them, so the model stays latent Gaussian.

Level 4: priors

parameter	role	prior
β_j, $\alpha_{j,r,y}$, δ_j, δ^{nb}_j, $\kappa_{\mathrm{U}\to\mathrm{C}}$, $\kappa_{\mathrm{C}\to\mathrm{U}}$, γ_r	fixed effects	$\mathcal{N}(0, 10^4)$
$\sigma_{u,j}$, $\sigma_{e,j}$, σ_a	SDs of iid effects	PC: $P(\sigma > 1) = 0.01$
$(\rho_{w,j}, \sigma_{w,j})$, (ρ_j, σ_j)	Matérn fields	PC: $P(\rho < 100\,\mathrm{km}) = 0.05$, $P(\sigma > 1) = 0.01$
$(\rho_\gamma, \sigma_\gamma)$	varying escalation	PC: $P(\sigma_\gamma > 0.2) = 0.05$
φ_j	persistence	$\operatorname{atanh} \varphi_j \sim \mathcal{N}(0, 1.5^2)$
λ_a, λ_r or ϑ_m	loadings	$\mathcal{N}(0, 2^2)$

Inference. Empirical Bayes: $\hat{\boldsymbol{\theta}}$ maximises the Laplace approximation of $p(\boldsymbol{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$; the latent effects and fixed effects are summarised by their Gaussian approximation at $\hat{\boldsymbol{\theta}}$; 95% intervals for $\boldsymbol{\theta}$ from the Hessian at $\hat{\boldsymbol{\theta}}$.

Special cases. $\lambda_a = 0$, $\sigma_a = 0$, $\lambda(s) \equiv \lambda$, $\gamma \equiv 0$, plus the neighbour cross terms: the current extended model. With $\lambda = \lambda_a = 0$: two separate univariate models.

Evidence so far (Africa and the Middle East; West Africa in brackets)

component	estimate, 95% interval	reading
δ_{C}	0.61 [0.59, 0.63] (0.45 [0.41, 0.48])	supported
$\kappa_{\mathrm{U}\to\mathrm{C}}$, no shared cell effect	0.094 [0.065, 0.123] (0.095 [0.032, 0.158])	supported
$\kappa_{\mathrm{U}\to\mathrm{C}}$, with shared static effect	0.057 [0.028, 0.086] (0.061 [−0.002, 0.124])	modest
neighbours' unrest (dropped)	−0.011 [−0.021, −0.001] (−0.005 [−0.021, 0.011])	about zero
lagged unrest, 13–26 weeks (dropped)	field diagnostic: corr. −0.04 (−0.08), placebo ≈ 0	no signal
shared static effect	log-ML +778 (+156); range 44 km (one cell)	supported; interval to do
λ	0.52 [0.46, 0.58] (0.58 [0.45, 0.71]); 0.35 with shared static	supported
φ_{C}, ρ_{C}	0.95 [0.945, 0.963], 453–551 km	supported
σ_{U}, φ_{U} (unrest only)	[0.40, 0.50], [0.54, 0.73]	supported
$\sigma_{w,\mathrm{U}}$ (static unrest field)	[0.29, 0.62] ([0.10, 1.23])	weak
$\gamma(s)$, $\lambda(s)$	—	not yet tested